Lab Manual

QuickMart Retail Billing System

Case Study

Scenario:
QuickMart is a small retail store that wants to automate parts of its daily billing and customer service operations. The store manager has asked you to build a simple console-based Python system to handle basic operations. Since the store doesn't have complex software, they need simple conditional logic.

The store needs help with:

- Calculating bills with discounts based on purchase amount
- Applying membership-tier benefits
- Categorizing customer age for special offers
- A simple menu system so staff can choose which operation to run

Q1. Discount Calculator (if-elif-else)

QuickMart gives discounts based on the total bill amount:

- Bill ≥ $5000 → 20% discount
- Bill ≥ $2000 → 10% discount
- Bill ≥ $1000 → 5% discount
- Below $1000 → No discount

Task: Take the bill amount as input, calculate the discount, and print the final payable amount.

```
Enter bill amount: 3500
Discount applied: 10%
Final amount to pay: 3150.0
```

Q2. Membership Tier Benefits (match-case)

QuickMart has 4 membership tiers, each with a different cashback percentage:

- "gold" → 15% cashback
- "silver" → 10% cashback
- "bronze" → 5% cashback
- "none" → 0% cashback

Task: Use match-case to take the membership tier and purchase amount as input, and print the cashback earned. Handle invalid tier input using the default case (case _:).

```
Enter membership tier: silver
Enter purchase amount: 2000
Cashback earned: 200.0
```

Q3. Age-Based Offer Eligibility (nested if-else)

QuickMart runs special offers based on age group:

- Age < 13 → "Kids Offer: Free toy on purchase above $500"
- 13 ≤ Age < 20 → "Teen Offer: 10% off on stationery"
- 20 ≤ Age < 60 → "Adult Offer: Standard discounts apply"
- Age ≥ 60 → "Senior Citizen Offer: Extra 15% discount"

Task: Take customer age as input and print the applicable offer. Additionally, check if the customer is also a member (yes/no input) - if yes, add "+ Priority Billing Counter" to the message.

```
Enter age: 65
Are you a member? (yes/no): yes
```

```
Senior Citizen Offer: Extra 15% discount + Priority Billing Counter
```

Q4. Staff Menu-Driven Billing Assistant (match-case, combines Q1–Q3 logic)

Build a simple menu so staff can choose which operation to perform:

```
1. Calculate Discount
2. Check Membership Cashback
3. Check Age-Based Offer
4. Exit
```

Task: Use match-case for the menu selection. Based on the choice, call the corresponding logic (you may reuse/simplify logic from Q1–Q3 as functions). Loop the menu until the user selects Exit.

```
1. Calculate Discount
2. Check Membership Cashback
3. Check Age-Based Offer
4. Exit
Enter your choice: 1
Enter bill amount: 6000
Discount applied: 20%
Final amount to pay: 4800.0

Enter your choice: 4
Thank you for using QuickMart Billing Assistant!
```